State of matter

Generally there are 4 states of matter

- (i) Solid
- (ii) liquid
- (iii) Gas
- (iv) Plasma

There are 6 types of bond

1. Ionic
2. Co-valent (H_2, H_2O) } mainly 3 bonds
3. Metalic (Cu, Fe) }
4. co-ordinate co valent bond
5. Hydrogen bond
6. Vander Waals forces

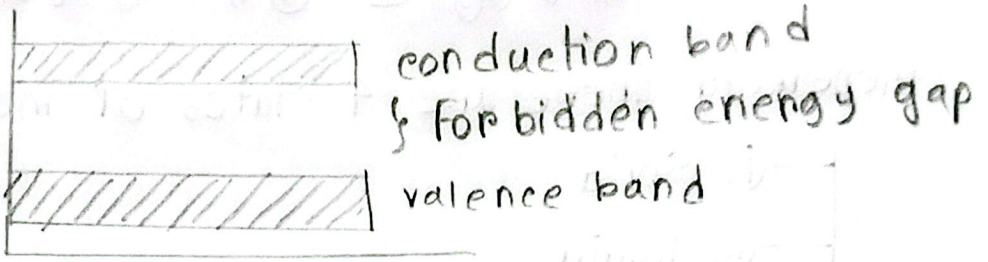
Band theory of matter:

Bond theory of matter is a ~~pro~~ theory that explains the electrical properties of solid materials based on the energy bands available to electrons and the energy gap between these bands.

It's important for understanding conductor, semi-conductor and Insulators.

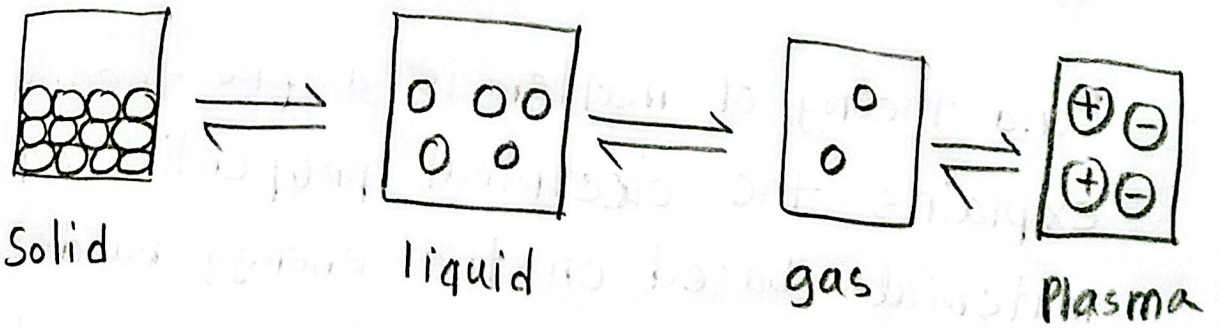
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Band theory of matter



অবগুলো atom এর কক্ষপথকে একত্রে করলে band তৈরি হয়।

Valence band থেকে conduction band এ electron আদান-প্রদানের উপর নির্ভর করে insulator, conductor, semi-conductor নির্ধারণ করা যায়।



Energy bands in solid

The ranges of energy's possessed by an electron in a solid known as energy band.

Valence band The range of energy's possessed by valence electrons is known as vallence band. This band may completely or partically filled.

For example: case of inert gas (Ne, He)

↓

In this case band is fully filled

Where in the other valence band are partically filled.

conduction band In certain materials the valence electrons are loosely attached in the cell. Even in a ordinary temperature some of the vallence electrons may get attached to become free electron. In fact, It is this free electrons which are responsible for the conduction of current in a conductor.

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for this reason these electrons are called conduction electron.

The range of possessed by conduction electron is known as conduction band.

All the electrons in conduction band are free electrons. Generally insulator have empty conduction band. On the other hand it is partially filled for conductors.

forbidden energy gap: The separation between valence band and conduction band on the energy level diagram is known as forbidden energy gap. Width of the forbidden energy gap measure of the bondage of valence electron. Greater the gap between them more tightly are bounded to the nucleous. In order to push electron to vallence band to conduction band to make the vallence electron free.

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External energy = forbidden energy gap must be supplied.

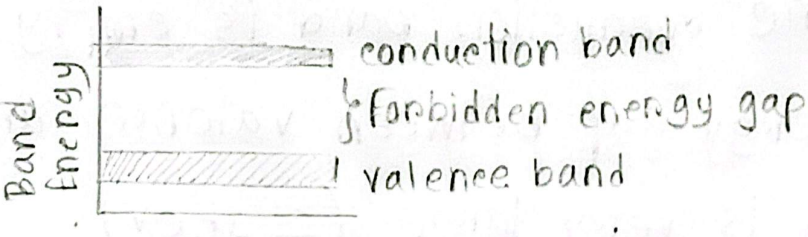


fig: 02

Classification of solids depands on the energy band

- conductor
- semi-conductor
- Insulator

The electroons of lower energy band are tightly bound to the nucleous and played no part in the conduction process. However the valence and conduction bands are of particular importance in assertaining the electrical behaviour of various solid.

(i) Insulator: Insulator are those substences which do not allow the passage of the electric

current flow them.

In terms of energy band, the valence band is full and the conduction band is empty.

Because the energy gap between valence and conduction band is very large ($\simeq$ 15eV).

Therefore at room temperature the vallence electrons of a insulators do not have enough to cross over to the conduction band.

However, when the temperature is raised, some of the valence electrons may gain enough energy to cross over the conduction band.

Hence, the registence of an insulator decreases with the increased in temperature

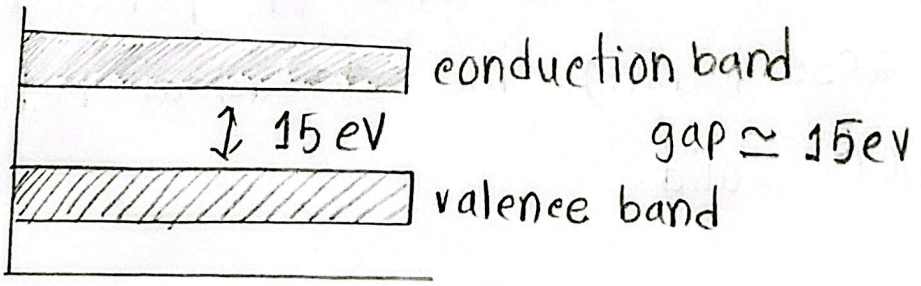


Fig 03 : Insulator

Example: Wood, glass, rubber etc

ii) Conductors: Conductors are those substance which easily allow the passage of electric current ~~allow~~ through them. It is because there are a large number of free electrons available in a conductor. In terms of energy band, the valence band and the conduction bands overlap each other. Due to this overlaping a slight potential difference accross a conductor causes the free electrons to constitute electric current. Thus electric behaviours of conductors can be satisfactorily explained by the energy theory of materials. In this there is no forbidden energy gap.

Example : Cu, Al

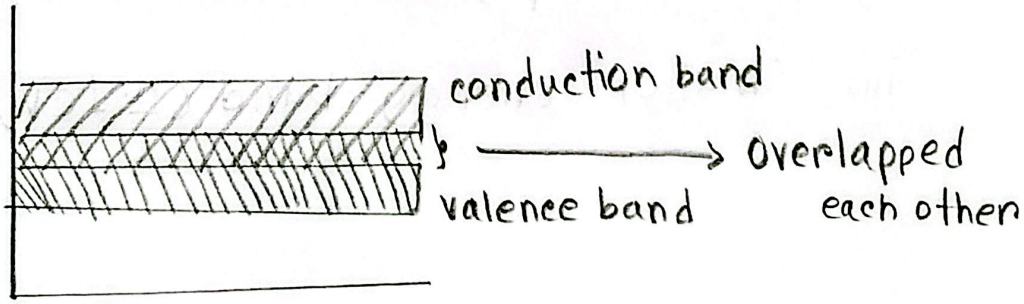


Fig : conductor

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(iii) Semiconductors: Semiconductors are those substances whose electrical conductivity lives in between conductors and insulators.

In terms of energy band, the valence band is almost filled and the th conduction band is almost empty. Further the energy gap between valence and conduction bands are very small. Therefore comparitively smaller electric field is required to push the electrons from the valence band to the conduction band.

In sumary semi-conductor has:

i) Almost full valence band

ii) Almost empty conduction band

iii) smaller energy band ($\simeq$ 1eV)

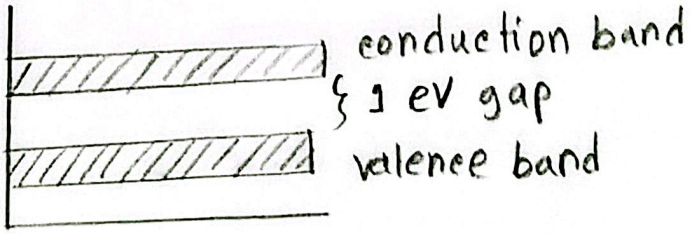


Fig : Semi-conductor

Semi-conductor in different temperature

At low temperature, a valence band is completely full. And conduction band is completely empty.

Therefore, a semi-conductor virtuely behaves as an insulator at low temperature.

As the temperature is increased more valence electrons cross over to the conduction band And the conductivity increases. It shows that electrical conductivity of semi-conductor increases with the rising temperature.

The primary states of matter

The primary states of matter are solid, ~~liqu~~ liquid, gas and Plasma. Which are ~~diso~~ distinguished by how a their component particles are arranaged and behave.

Solid: Tightly packed particles in fixed positions, provide a definite shape and volume. The interatomic distance between the atoms is very small.

Liquid: close particles that flow-past each other maintain a fixed volume while taking the shape of the container.

Gas: Widely spaced, freely moving particles expand to fill the entire container. The inter atomic distance between the atoms is very large.

Plasma: High energy ionized gas with free electrons, influenced by electro magnetic fluids. They found naturally in lightening and stars.

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The electrical conductivity of a semi-conductor changes with temperature variation

(i) At absolute zero: At absolute zero temperature, the electrons are tightly held by the semi-contactors atoms.

The inner orbit electroons are bound ~~Whea~~ whereas the valence electroons are engaged in co-valent bindingly.

At this temperature, the co-valent bonds are very strong. And there are no free electrons. so, the semiconductor crystals behave as a perfect insulator.

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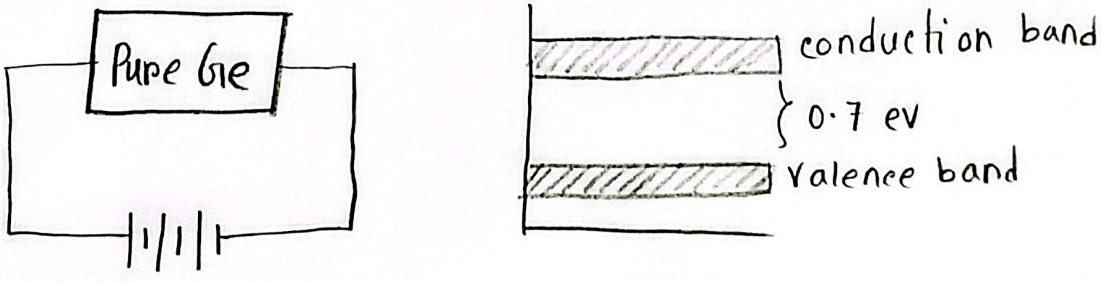


Fig : Gie at absolute zero degree

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(ii) Above Absolute zero: When the temperature is raised some of the co-valent bonds in the semi-conductor break due to the thermal energy supply. The breaking of bonds sets those electroons free. which are engaged in the formation of those bonds. The result is that, A few free electroons exists in the semi-conductor. These free electroons can constitute a tiny electric current. If potential difference is applied accross the semi-conductor. These shows that the registence of a semi-conductor decrease with the rising temperature.

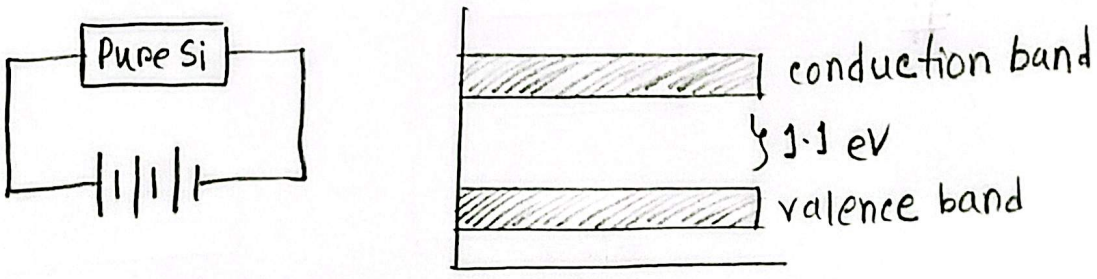
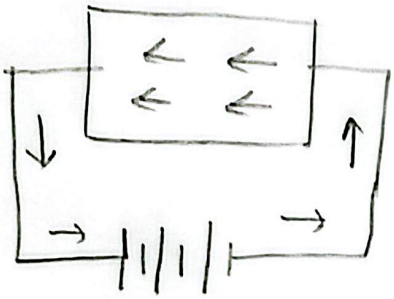
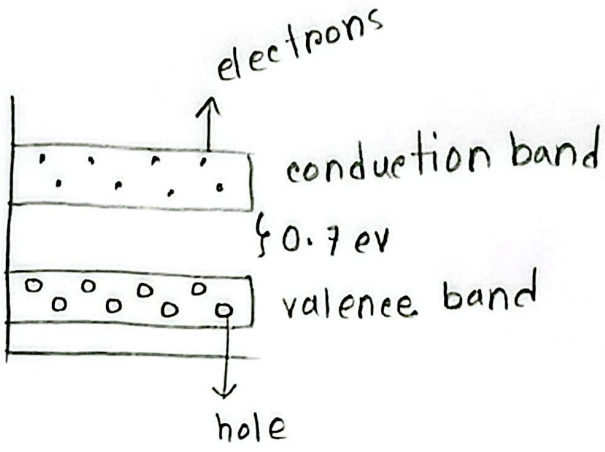


Fig : Si at above absolute zero

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Hole: When co-valent is broken, the removal of one electron leaves a vacancy, a missing electron in the co-valent bond. These missing electroons are called hole. Which acts as a positive change.





Fig

Classification of semi-conductors

- Intrinsic
- Extrinsic